Suspended Sediment Particle Size Prediction

Random Forest, Support Vector Regression, and XGBoost

Problem statement

This project compares three regression models for estimating median particle diameter (d50) and particle-size-distribution variance (sigma2) from environmental predictors in the supplied synthetic dataset. The report follows the original step-by-step notebook and reproduces its saved comparison table.

Dataset and preparation

The supplied *synthetic_data_improved.csv* contains 1,648 hourly rows, eight predictors, and two target columns. The time column and incomplete rows were removed, leaving 1,402 rows. The notebook then removed rows at or above the 99th percentile of either target, leaving 1,384 rows, and shuffled the remaining data into a 70% training set (968 rows) and a 30% test set (416 rows) using random_state=42.

Approach and implementation

A Random Forest importance table was fit using the training portion of d50. The notebook then manually fixed the model inputs to S, ub, np, and T. Random Forest used 400 trees. SVR used standardized inputs and a five-fold grid search over C and epsilon. XGBoost used 300 trees, depth 3, and learning rate 0.05. Each model was scored on the held-out test portion using R^2, RMSE, and MAE. The notebook and script are included with the project materials.

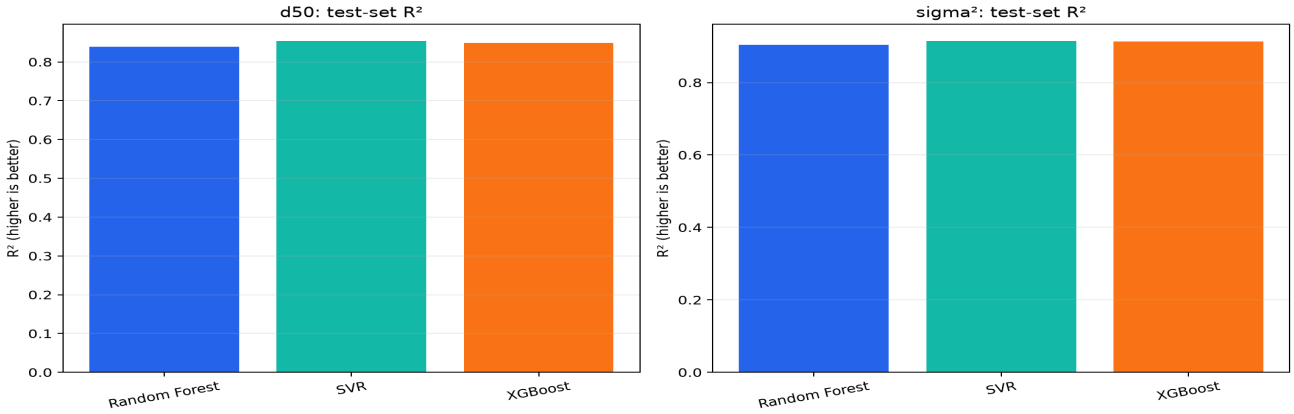
Notebook-reported test results

Target	Model	R^2	RMSE	MAE
d50	Random Forest	0.839729	15.769309	12.557175
sigma2	Random Forest	0.904587	7.931298	6.251205
d50	SVR	0.854116	15.044842	11.969561
sigma2	SVR	0.914782	7.495579	5.801557
d50	XGBoost	0.849638	15.274025	12.284159
sigma2	XGBoost	0.913948	7.532165	5.880333

Higher R^2 and lower RMSE/MAE indicate better fit on this particular test split.

Findings and limitations

Results from the saved notebook outputs



Conclusion

SVR recorded the highest R^2 for both targets in the saved notebook results: 0.854116 for d50 and 0.914782 for sigma2. Random Forest scored 0.839729 and 0.904587; XGBoost scored 0.849638 and 0.913948, respectively. These scores describe the filtered synthetic dataset and the notebook's random holdout.

Evaluation limitations

The notebook calculates target percentiles using all complete rows before making the train/test split. Since the test labels affect which rows remain in the dataset, this can bias the reported scores upward. The score values should be presented as notebook-reported results, not as an unbiased estimate of future or field performance. The data are synthetic, and a shuffled split does not evaluate future-period forecasting.

Feature list note

The saved importance table ranks ub, S, T, and chl_a highest. The notebook manually sets the model feature list to S, ub, np, and T, so the selected list does not exactly match the ranking. The supplied notebook is preserved unchanged; the reported scores use its manually specified feature list.

Reproducibility

The project includes the original notebook, the supplied dataset, requirements.txt, setup instructions, a script that mirrors the notebook, and saved outputs. Install the dependencies with `pip install -r requirements.txt` and run `python run_project.py` from the project root. Small score differences can occur across software versions.

Team members: _____ _____